



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Optimisation & Monte Carlo Sampling

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

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Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

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- *Without Prior Knowledge*: out of n independent measurements the outcome A occurs k times, defining

$$P(A) = \lim_{n \rightarrow \infty} \frac{k}{n}$$

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Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

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■ *Mean* or first raw moment: $\mu = E[X] = \int_{-\infty}^{+\infty} x f(x) dx$

■ *Variance* or second raw moment: $\sigma^2 = E[(X - E[X])^2] = \int_{-\infty}^{+\infty} (x - E[x])^2 f(x) dx$

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- *Arbitrary Distributions:* obtained from uniformly distributed numbers by various methods

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$$I_{\nu}(T, \beta) = B_{\nu}(T) (1 - e^{-\tau_{\nu}(\beta)}) \approx B_{\nu}(T) \tau_{\nu}(\beta) = B_{\nu}(T) \Sigma \kappa_{\nu}(\beta) = \frac{2h\Sigma\kappa_0}{c^2\nu_0^{\beta}} \frac{\nu^{3+\beta}}{\exp\left(\frac{h\nu}{kT}\right) - 1}$$

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 - suppose we have no prior information on the parameter distribution

Example

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■ *Likelihood:*

$$L(I_v | v, \sigma_v, \theta) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_{v,i}^2}} \exp\left(-\frac{(I_{v,i} - I_v(\theta))^2}{2\sigma_{v,i}^2}\right)$$

Example

■ *Likelihood:*

$$\log L(I_v | v, \sigma_v, \theta) = -\frac{N}{2} \log(2\pi) - \sum_{i=1}^N \sigma_{v,i} - \frac{1}{2} \sum_{i=1}^N \frac{(I_{v,i} - I_v(\theta))^2}{\sigma_{v,i}^2}$$

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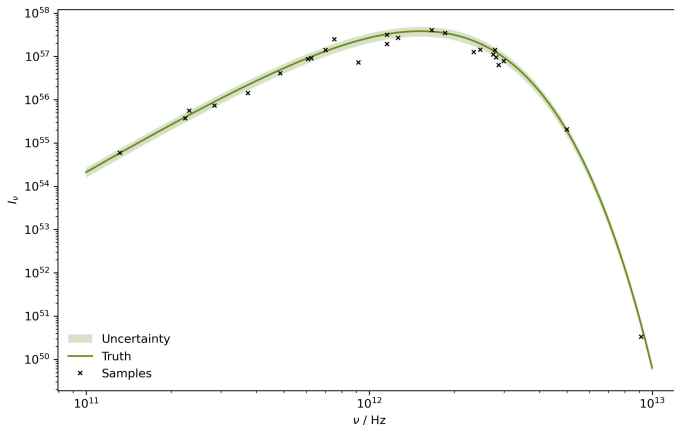
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- *Objective:* minimise the $\chi^2(\theta)$ measure to maximise the posterior probability

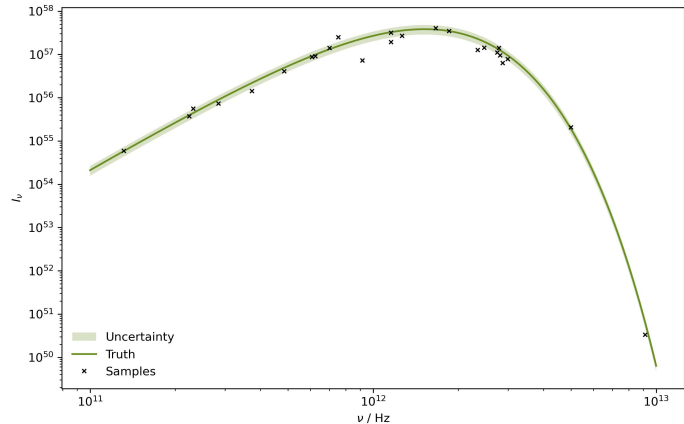
Measurements

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How do we proceed from this dataset to maximise the likelihood for our nonlinear model?



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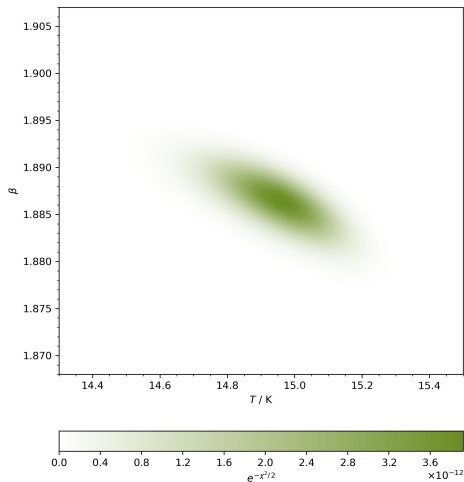
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- *Problem:* as D increases we have N^D gridpoints, and with more complex parameter distributions, this procedure quickly becomes intractable due to computational limitations

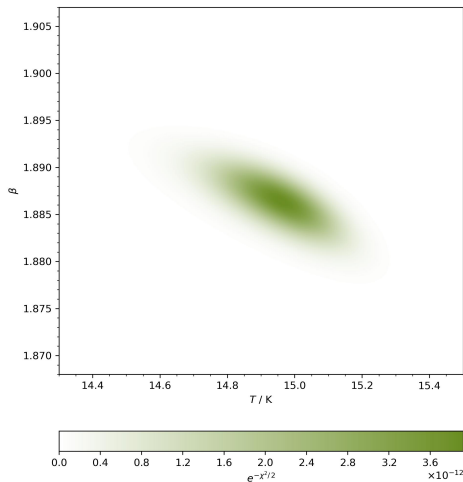
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Can we think of more elegant ways to explore even very high dimensional parameter spaces?



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- *Interactive Visualisations*: check out Ben Frederickson¹ and Andy Casey² to see some nice examples

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 - zeroth order method since no derivatives are used

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Nelder-Mead

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Nelder-Mead

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- ▶ if x_r better than second worst x_d but not better than best x_1 with $f(x_1) \leq f(x_r) < f(x_d)$ then obtain new simplex by replacing x_{d+1} with x_r and repeat from start

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- ▶ continue to the next step instead if $\mathbf{x}_r$ is the best point with $f(\mathbf{x}_r) < f(\mathbf{x}_1)$

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- ▶ continue to the next step instead if $\mathbf{x}_r$ is the best point with $f(\mathbf{x}_r) < f(\mathbf{x}_1)$
- ▶ skip the next skip and go to the one after it if $\mathbf{x}_r$ is not better than the second worst $\mathbf{x}_d$ with $f(\mathbf{x}_r) \geq f(\mathbf{x}_d)$

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- ▶ compute expanded point $x_e = x_0 + \gamma(x_r - x_0)$ with $\gamma > 1$
- ▶ if x_e is better than x_r with $f(x_e) < f(x_r)$ then obtain new simplex by replacing the worst x_{d+1} with x_e and repeat from start
- ▶ else obtain new simplex by replacing the worst x_{d+1} with x_r and repeat from start

3. Contraction:

- ▶ if x_r is better than the worst x_{d+1} with $f(x_r) < f(x_{d+1})$ then compute the outside contracted point $x_c = x_0 + \rho(x_r - x_0)$ where $0 < \rho \leq 0.5$

Nelder-Mead

■ Conditions:

2. Expansion:

- ▶ compute expanded point $x_e = x_0 + \gamma(x_r - x_0)$ with $\gamma > 1$
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- ▶ for x_c better than x_r with $f(x_c) < f(x_r)$ obtain new simplex by replacing worst x_{d+1} with x_c and repeat from start, otherwise go to the next step

Nelder-Mead

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- *Conditions:*

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- 3. *Contraction:*

Nelder-Mead

■ Conditions:

3. Contraction:

- ▶ if x_r is not better than the worst x_{d+1} with $f(x_r) \geq f(x_{d+1})$ then compute the inside contracted point $x_c = x_0 + \rho(x_{d+1} - x_0)$ where $0 < \rho \leq 0.5$

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4. Shrink:

Nelder-Mead

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4. Shrink:

- ▶ obtain new simplex by replacing all points except the best x_1 with $x_i = x_1 + \sigma(x_i - x_1)$ and coefficient $0 < \sigma < 1$ before repeating from start

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- ▶ if x_r is not better than the worst x_{d+1} with $f(x_r) \geq f(x_{d+1})$ then compute the inside contracted point $x_c = x_0 + \rho(x_{d+1} - x_0)$ where $0 < \rho \leq 0.5$
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■ Standard Values: $\alpha = 1$ & $\gamma = 2$ & $\rho = \sigma = 0.5$

Random Descent

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- *Requirements:* only objective function, no information on gradients

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 4. accept proposal $x_{k+1} = \hat{x}_{k+1}$ only if $f(\hat{x}_{k+1}) < f(x_{k+1})$ and repeat

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- *Notes:*
 - traverses parameter space via modified random walk, can be very slow
 - zeroth order method since no derivatives are used
 - improve by considering last step and optimizing step range

Gradient Descent

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- *Requirements:* first derivatives of the objective function, can be estimated numerically

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 1. initialize at some x_0

Gradient Descent

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- *Algorithm:*
 1. initialize at some x_0
 2. update current point with $x_{k+1} = x_k - \alpha \nabla f(x_k)$

Gradient Descent

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- *Algorithm:*
 1. initialize at some x_0
 2. update current point with $x_{k+1} = x_k - \alpha \nabla f(x_k)$
 3. iterate until convergence $|x_{k+1}/x_k - 1| < \epsilon$ or maximum step count is reached

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- *Notes:*
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 - improve by performing line search for α at every step and making sparse updates

Backtracking Line Search

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Backtracking Line Search

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- *Algorithm:*
 1. start from some fixed large α_0 for each step k of the gradient descent
 2. set $\alpha_{j+1} = \tau \alpha_j$ until $f(x_k) - f(x_k - \alpha_j \nabla f(x_k)) \geq \kappa \alpha_j (\nabla f(x_k))^2$ where $\tau, \kappa \in (0, 1)$

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- *Notes:*
 - ensures $f(x_{k+1}) < f(x_k)$ such that the function steadily decreases with increasing k

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 - first order method since only the gradient is used
 - typical settings are $\tau = \kappa = 0.5$ (L. Armijo, *Pac. J. Math.* 1966)

Conjugate Gradient Descent

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■ *Algorithm:*

1. calculate search direction $s_k = \nabla f(x_k) + \beta s_{k-1}$ where $\beta = \frac{\nabla f(x_k)(\nabla f(x_k) - \nabla f(x_{k-1}))}{(\nabla f(x_{k-1}))^2}$

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■ *Notes:*

- uses gradients from previous evaluations to estimate curvature

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- smooths oscillatory behaviour of naive gradient descent, especially in tight valleys with nearby gradients almost orthogonal to search direction

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Adaptive Moment Estimation

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Adaptive Moment Estimation

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- *Algorithm*:
 1. initialize position $\mathbf{x}_0$ as well as moving averages $\mathbf{m}_0 = \mathbf{0}$ and $\mathbf{v}_0 = \mathbf{0}$

Adaptive Moment Estimation

- *Requirements*: first derivatives of the objective function, can be estimated numerically
- *Algorithm*:
 1. initialize position x_0 as well as moving averages $m_0 = 0$ and $v_0 = 0$
 2. set $m_{k+1} = \beta_1 m_k + (1 - \beta_1) \nabla f(x_k)$ and $v_{k+1} = \beta_2 v_k + (1 - \beta_2) (\nabla f(x_k))^2$

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 3. correct estimator bias with $\tilde{m}_k = m_{k+1} (1 - \beta_1^k)^{-1}$ and $\tilde{v}_k = v_{k+1} (1 - \beta_2^k)^{-1}$

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3. correct estimator bias with $\tilde{m}_k = m_{k+1} (1 - \beta_1^k)^{-1}$ and $\tilde{v}_k = v_{k+1} (1 - \beta_2^k)^{-1}$
4. update the position $x_{k+1} = x_k - \alpha \tilde{m}_k (\sqrt{\tilde{v}_k} + \epsilon)^{-1}$ and repeat

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■ *Notes*:

- point behaves as iterative analogue to ball rolling in landscape with friction

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- hyperparameters $\beta_1, \beta_2 \in (0, 1)$ are forgetting factors for previous gradients, and $\epsilon \ll 1$

Adaptive Moment Estimation

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Newton Method

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- *Requirements:* first and second derivatives of the objective function, can be estimated numerically

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- *Algorithm:*
 1. choose some initial x_0

Newton Method

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- *Algorithm:*
 1. choose some initial x_0
 2. calculate $\nabla f(x_k) = \left(\frac{\partial f(x_k)}{\partial z_i} \right)_{i=1,\dots,d}$ and $H_{f(x_k)} = \left(\frac{\partial^2 f(x_k)}{\partial z_i \partial z_j} \right)_{i,j=1,\dots,d}$

Newton Method

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- related Gauss-Newton algorithm employs Jacobian instead of Hessian

Advanced Algorithms

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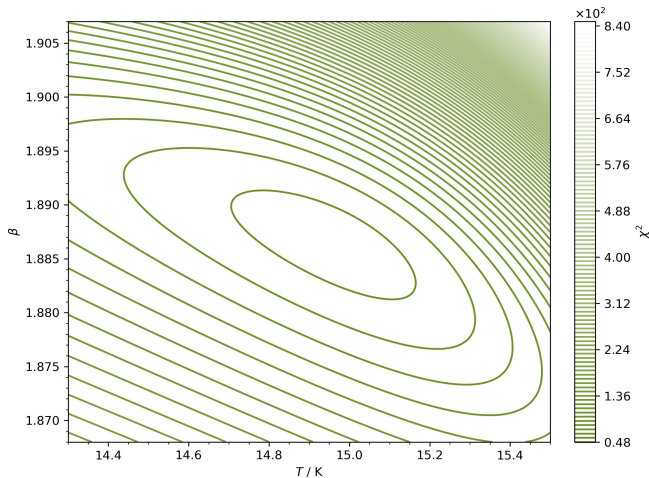
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- *Usage*: found in many modern libraries, suited to specialised use cases like nonlinear least squares, problems with parameter bounds or arbitrary constraints

Application

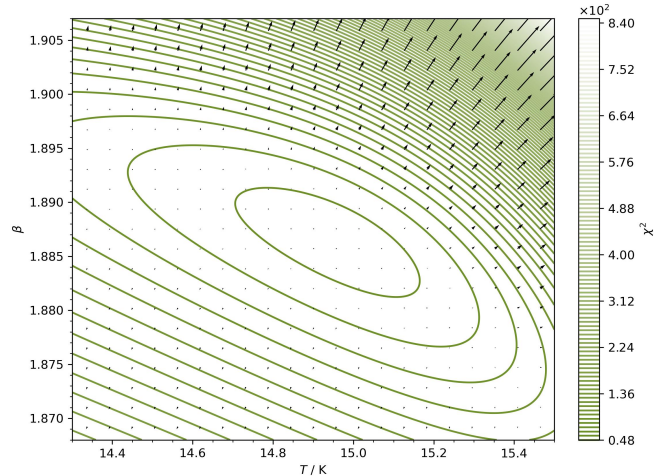
Application

- Example: minimise $\chi^2(\theta)$ using the gradient descent algorithm



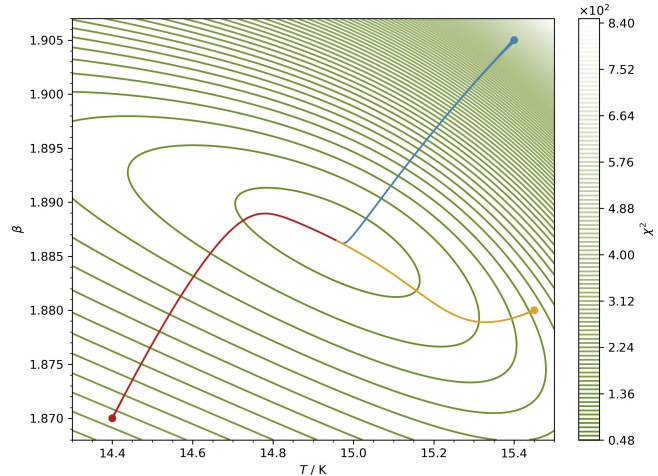
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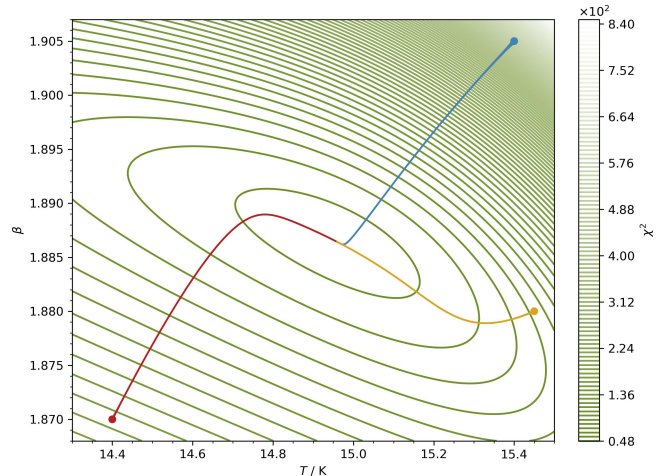
Application

- Example: minimise $\chi^2(\theta)$ using the gradient descent algorithm
- Fixed Rate: $\alpha = 10^{-9}$



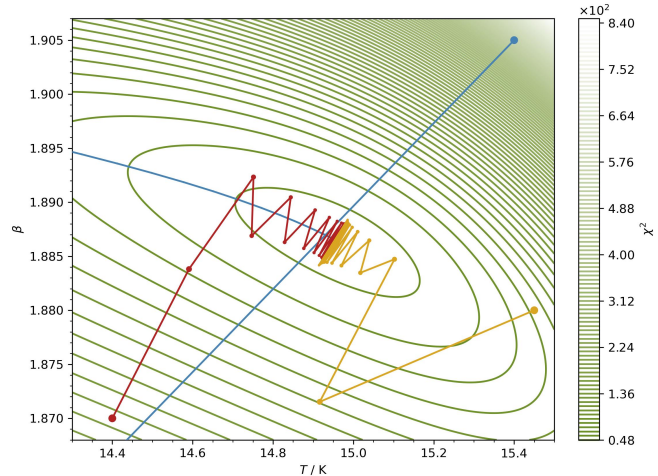
Application

- *Example:* minimise $\chi^2(\theta)$ using the gradient descent algorithm
- *Fixed Rate:* $\alpha = 10^{-9}$
 - very smooth, but does not converge, step size too small



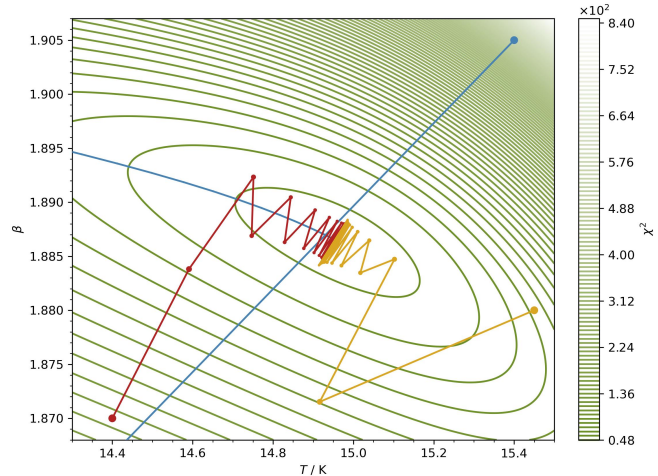
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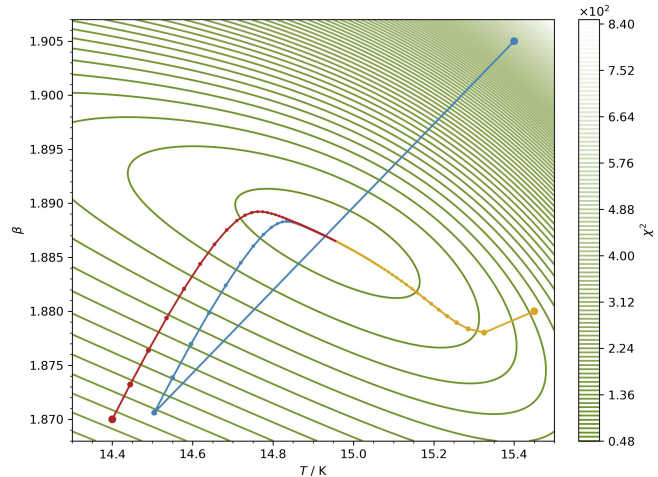
Application

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 - oscillations, does not converge, step size too large



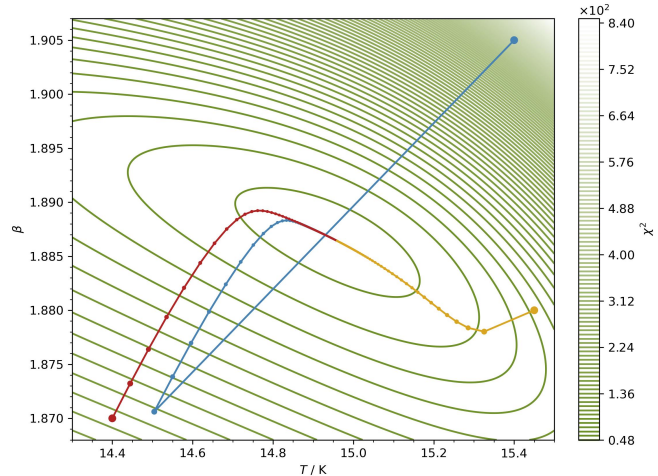
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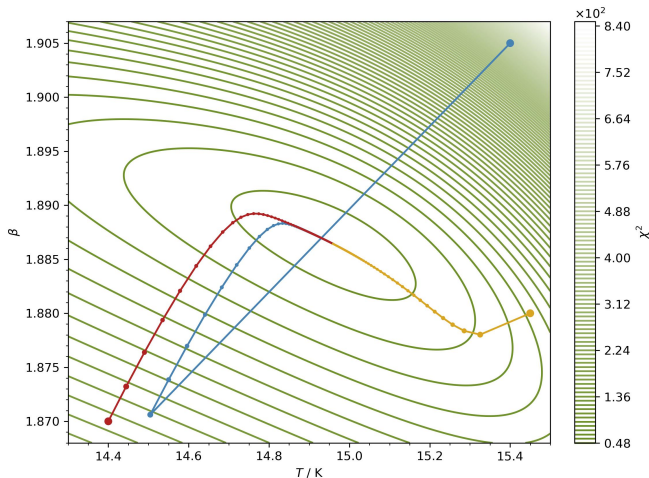
Application

- *Example:* minimise $\chi^2(\theta)$ using the gradient descent algorithm
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 - this looks good, optimisers converge



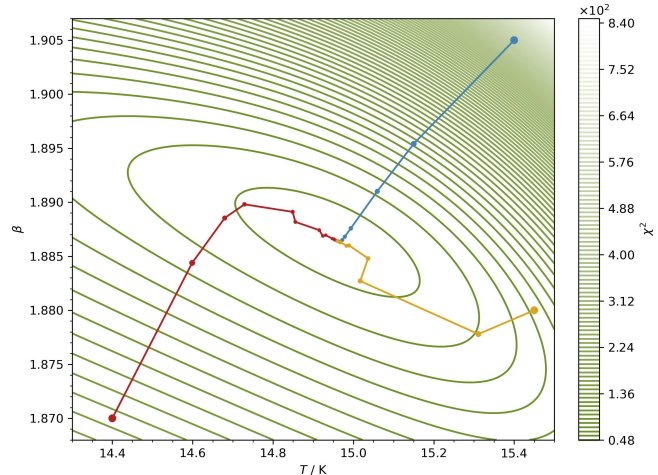
Application

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 - $T_{400} = 14.96 \text{ K}$ & $\beta_{400} = 1.866$



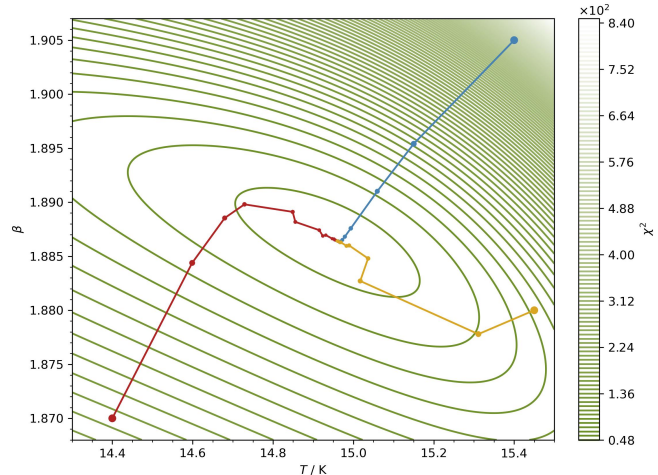
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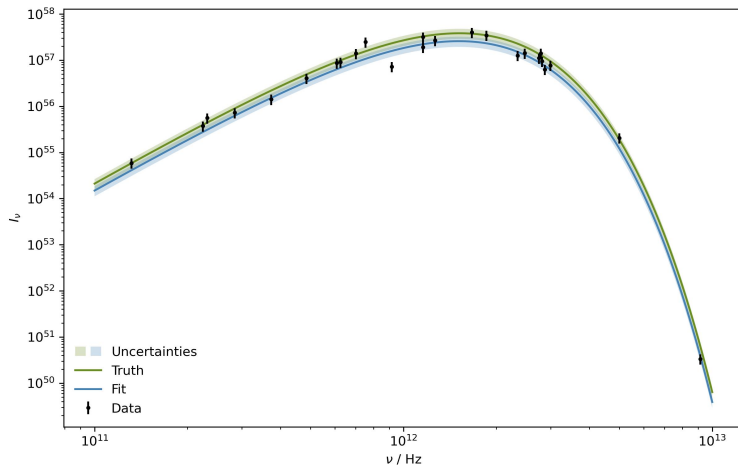


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Result



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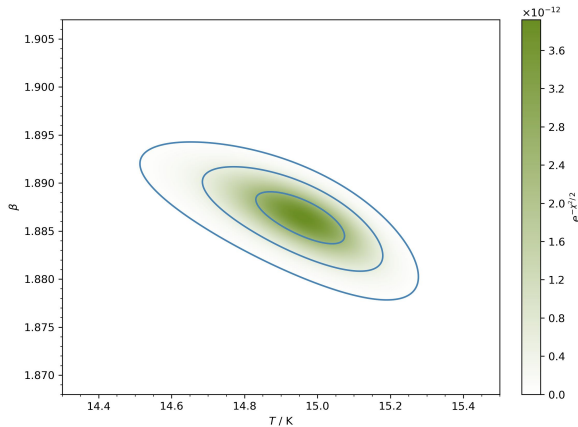
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Alternatives

Estimating Covariance

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■ Gridded Posterior Contours

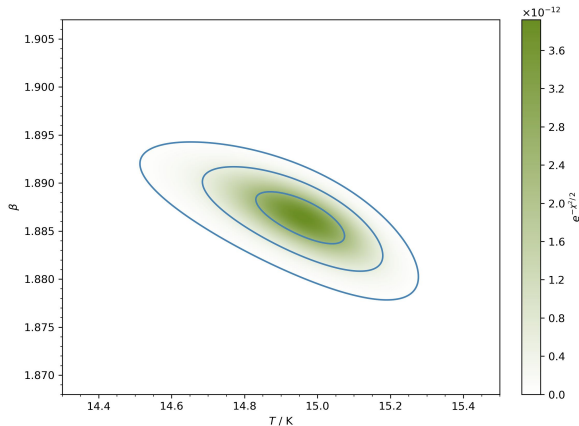


Estimating Covariance

■ *Gridded Posterior Contours*

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$$x \sim N(\mu, \Sigma) \propto \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$$



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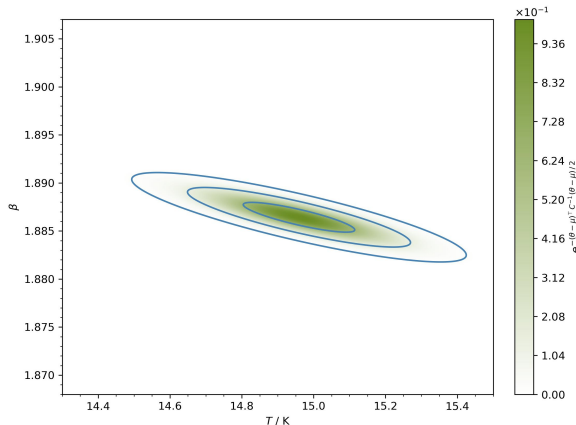
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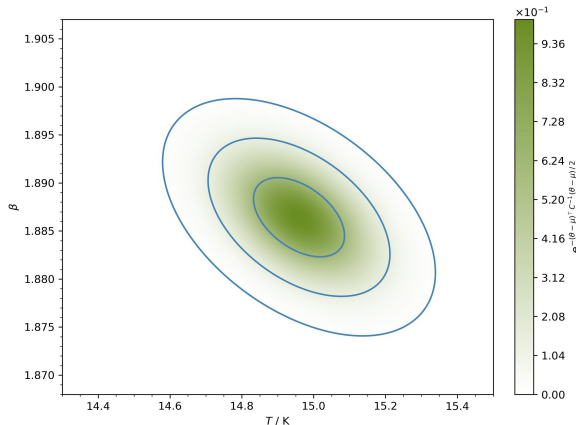
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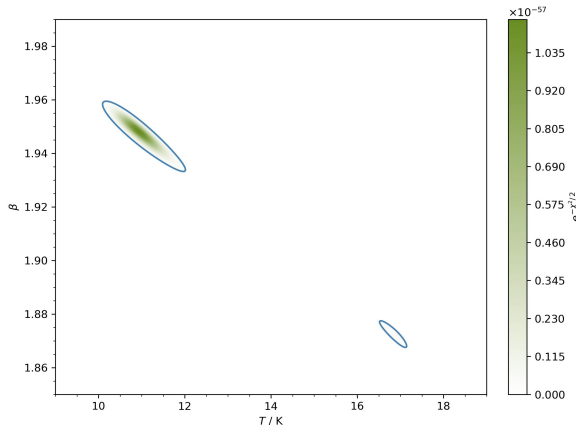
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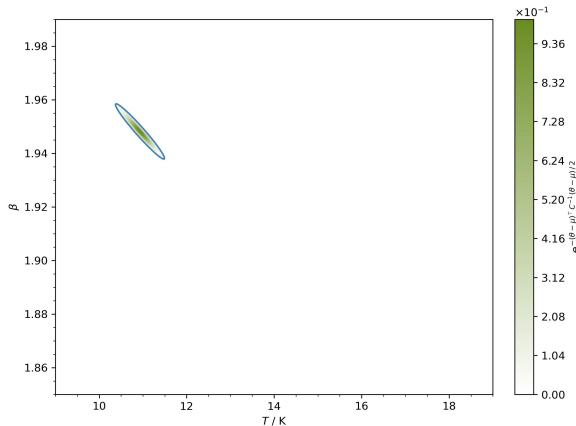
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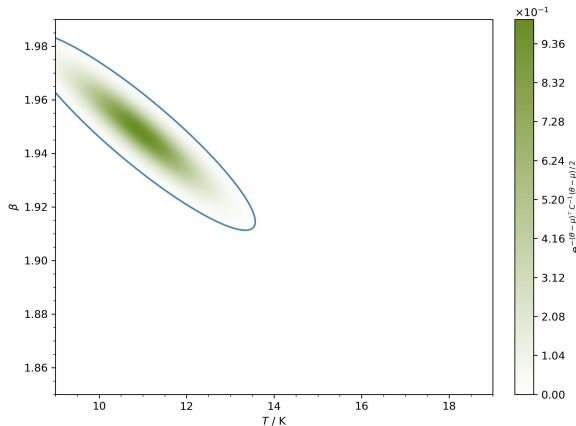
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MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



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- fixing some parameters or optimising independently are potential solutions



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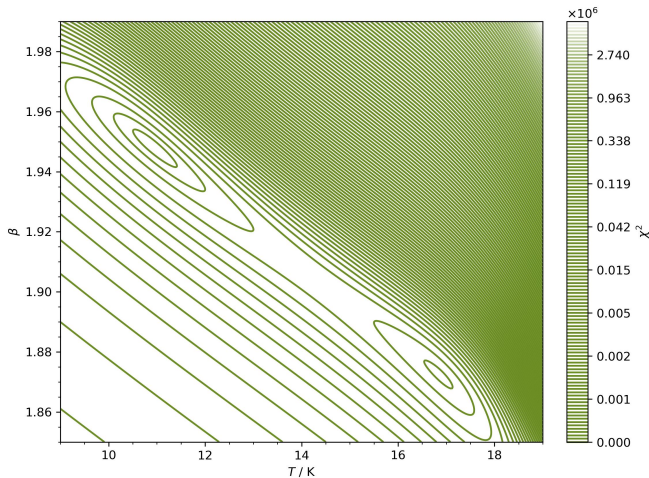


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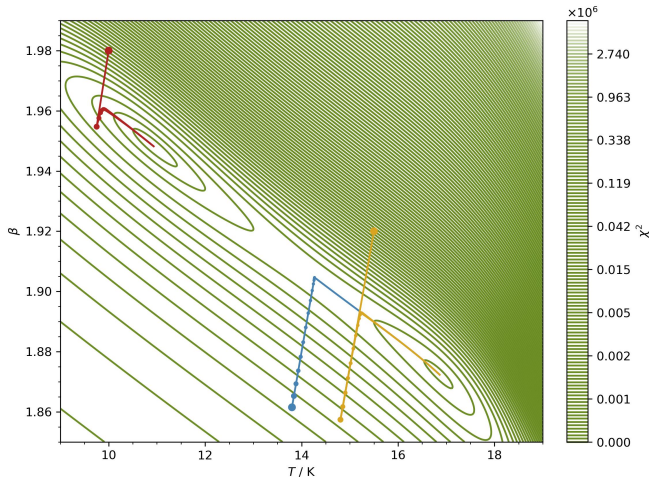
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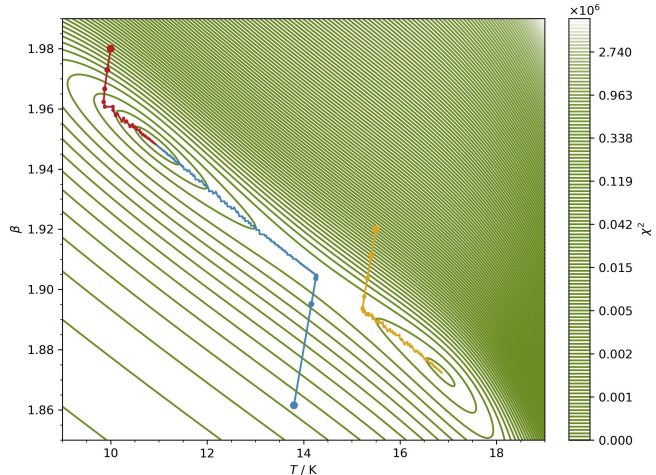
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Limitations & Curses

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 - Line Search Gradient Descent:
 $T_{660} = 10.93 \text{ K}$ & $\beta_{660} = 1.948$



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 - estimate expectation values by integrating
 - uncertainty propagation to the model function
 - simulate complex measurement processes and quantify the total convolution

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- then the average behaviour of this single process reproduces that over the entire ensemble

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- unlike the underlying posterior that we sample, the proposal must be properly normalized

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- a proposal x_j is drawn from a fixed step size distribution such as a normal or uniform one that is symmetric in each coordinate $g(x_j|x_i) = g(x_i|x_j)$
- this satisfies detailed balance $M_{i \rightarrow j} f(x_i) = M_{j \rightarrow i} f(x_j)$ with

$$M_{i \rightarrow j} = \min \left(1, \frac{f(x_j)}{f(x_i)} \right)$$

Markov Chain Monte Carlo

■ Metropolis:

- a proposal x_j is drawn from a fixed step size distribution such as a normal or uniform one that is symmetric in each coordinate $g(x_j|x_i) = g(x_i|x_j)$
- this satisfies detailed balance $M_{i \rightarrow j} f(x_i) = M_{j \rightarrow i} f(x_j)$ with

$$M_{i \rightarrow j} = \min \left(1, \frac{f(x_j)}{f(x_i)} \right)$$

- generate a random number $\xi \sim U(0, 1)$ and decide according to

$$x_{i+1} = \begin{cases} x_j & \xi \leq M_{i \rightarrow j} \\ x_i & \xi > M_{i \rightarrow j} \end{cases}$$

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- asymmetric proposals allow efficient handling of bounded parameter spaces
- proposal distribution shapes can be tailored to the sampled posterior to speed up exploration of the space and achieve good convergence
- realistic cases, especially when priors are included, are generally not symmetric or unbounded, therefore benefitting from the inbuilt bias correction to ensure correct sampling of the target distribution

Advanced Methods

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- *Affine-Invariant:*

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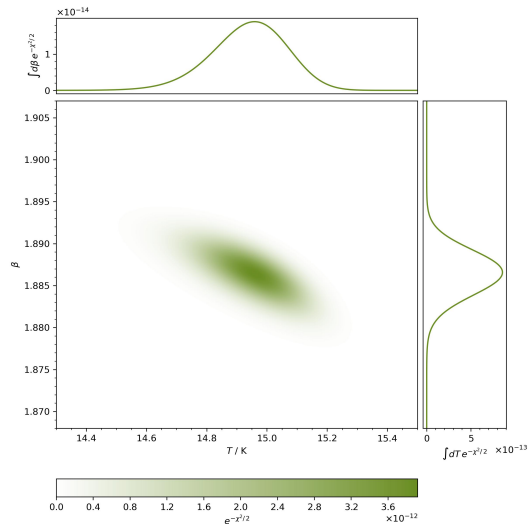
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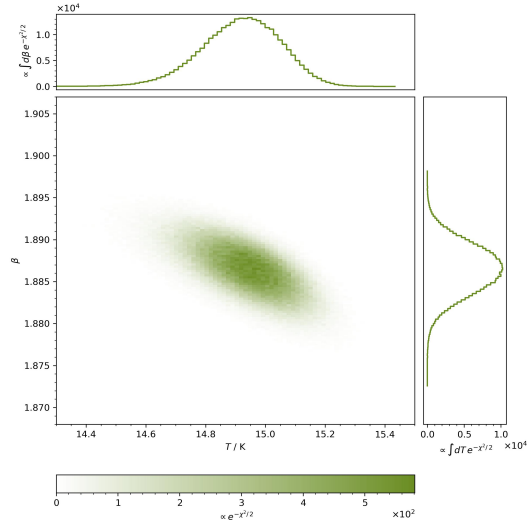
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- efficient for many dimensions with very high acceptance ratios

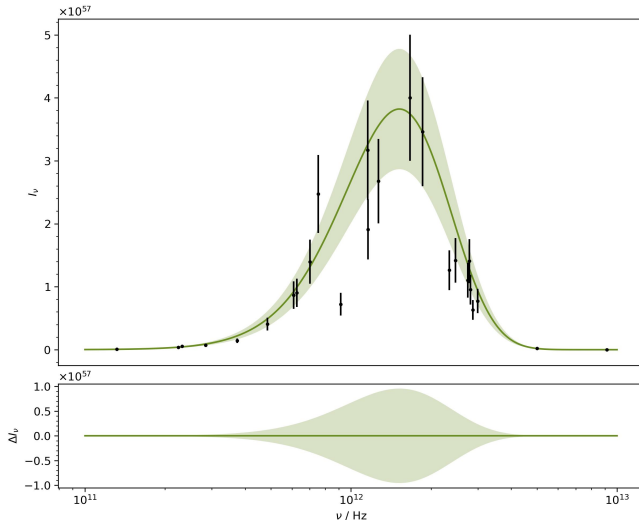
Posterior



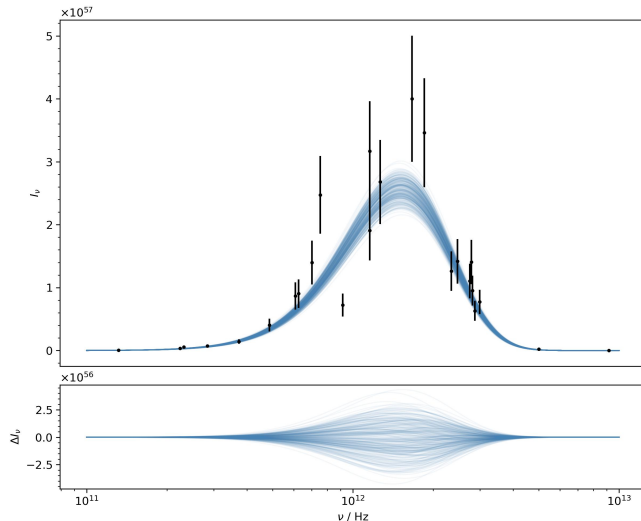
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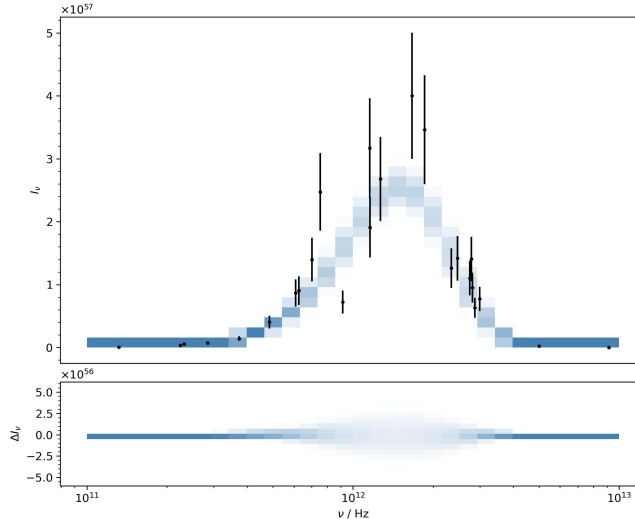
Comparison



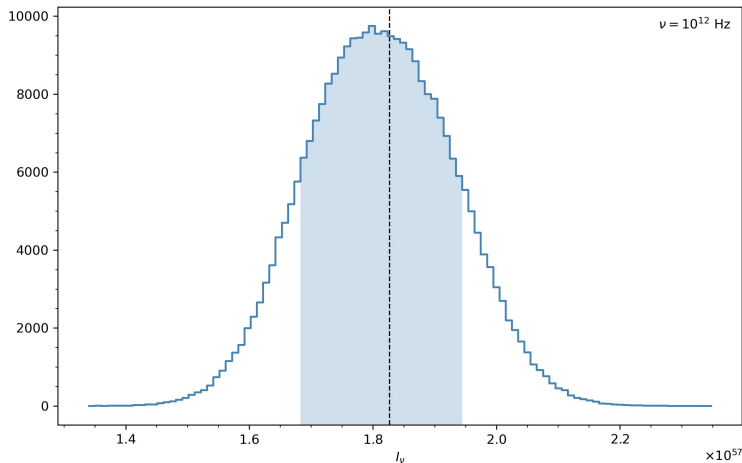
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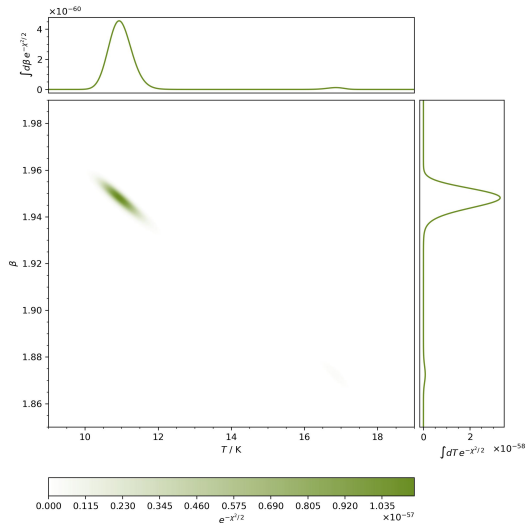
Comparison



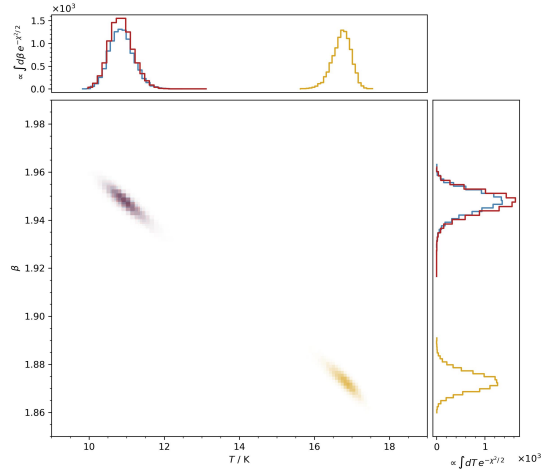
Error Propagation



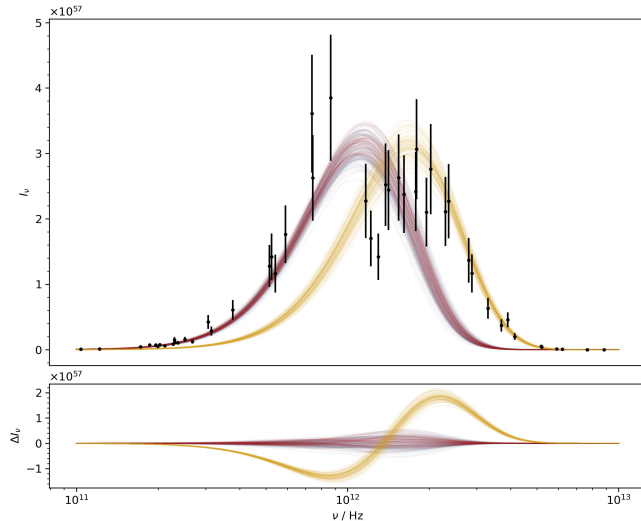
Posterior



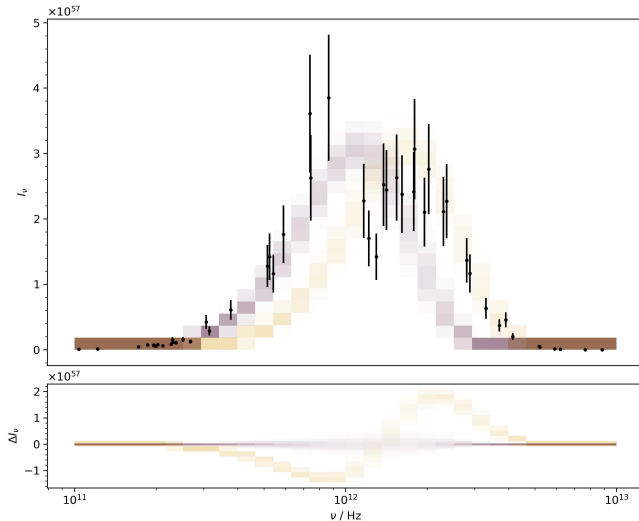
Posterior



Comparison



Comparison



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- identify burn in phase via trace plot and reject it

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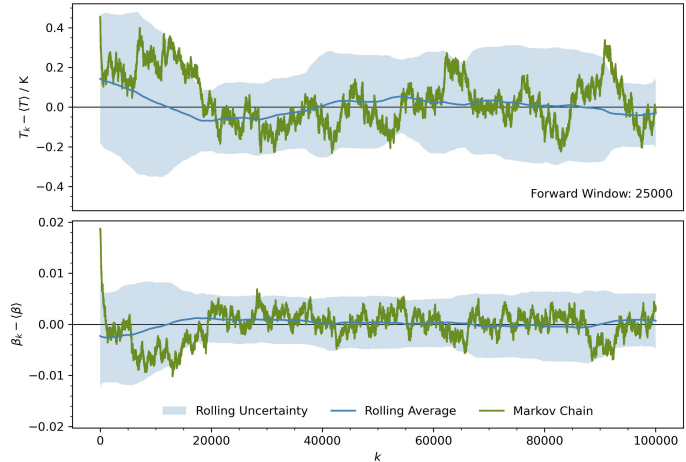
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- autocorrelation length l is defined by lag t after which $|\rho_t| < \sigma_t$ and therefore $\rho_t \neq 0$ no longer statistically significant

Application

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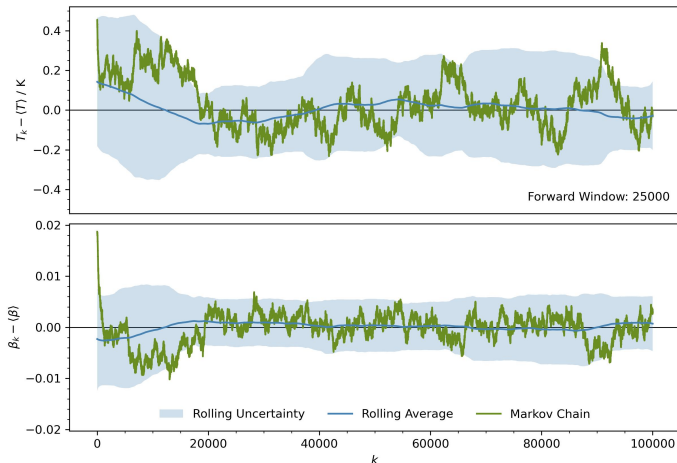
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■ $h = 10^{-4}$ & acc = 97.72%

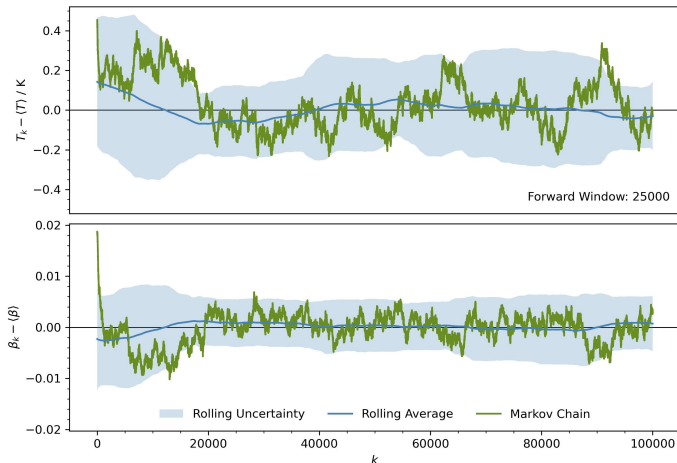


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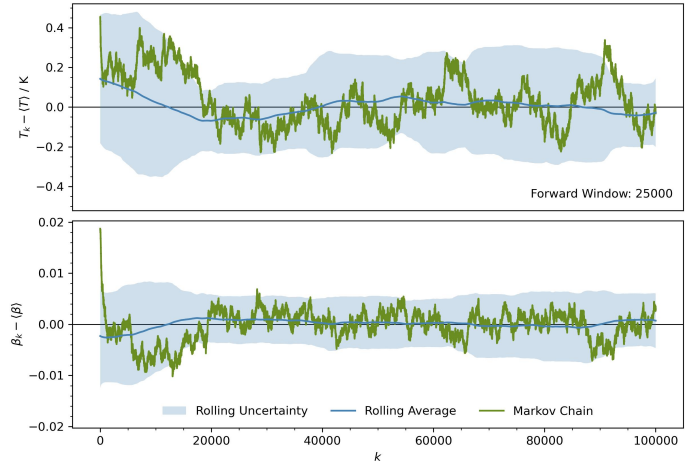
■ acceptance too high



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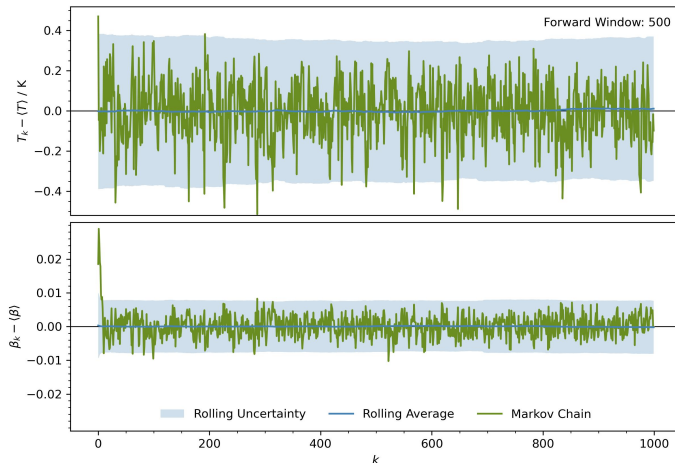
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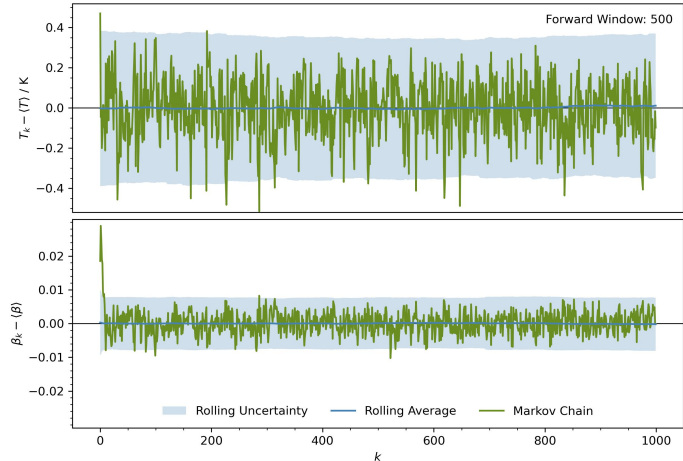
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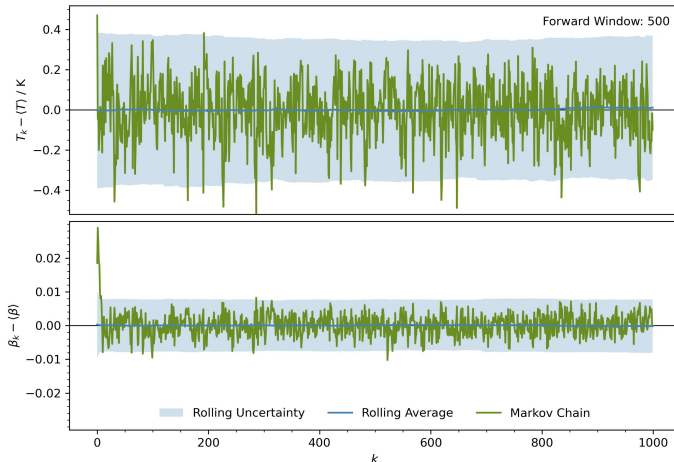


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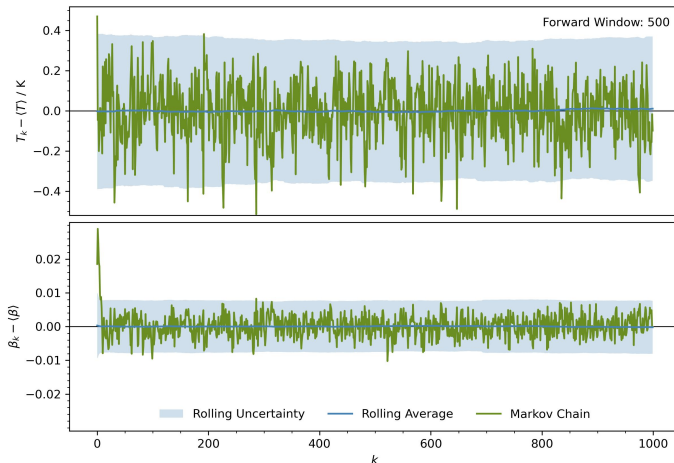
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Application

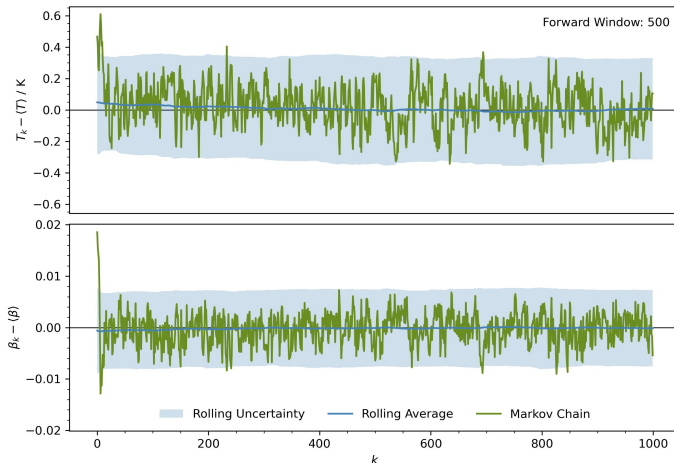
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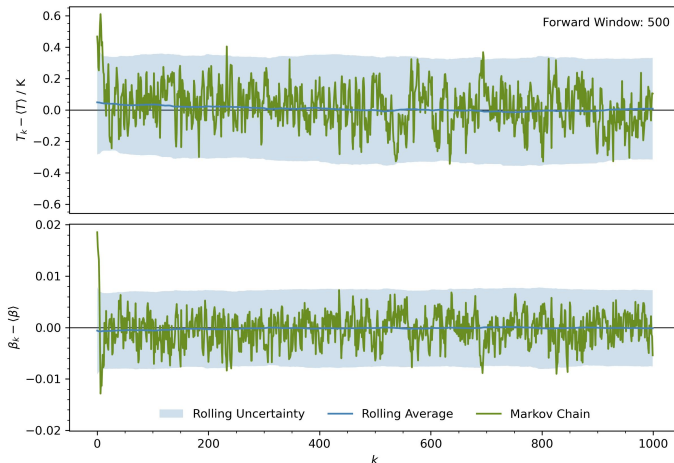
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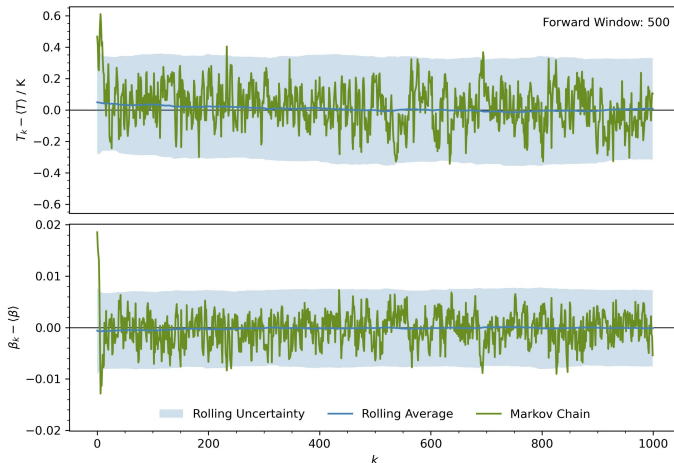


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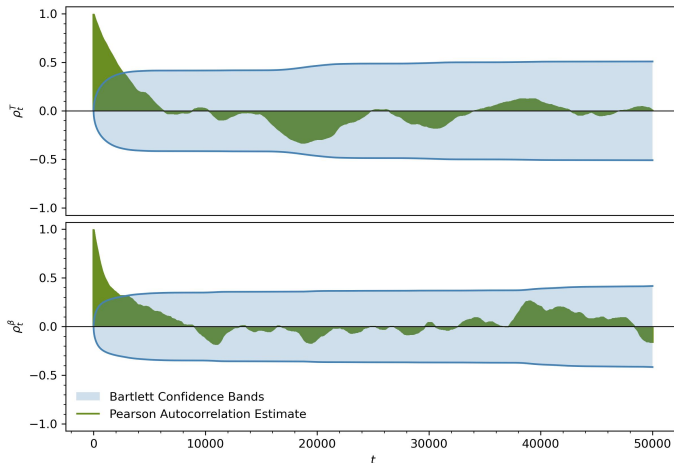
■ acceptance and step size
seem good



Application

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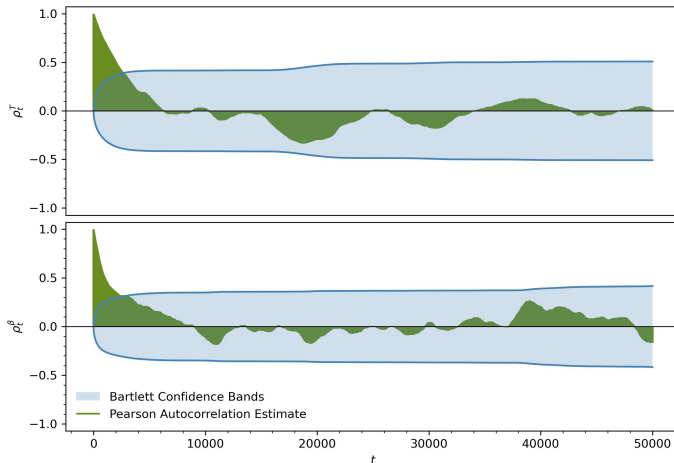
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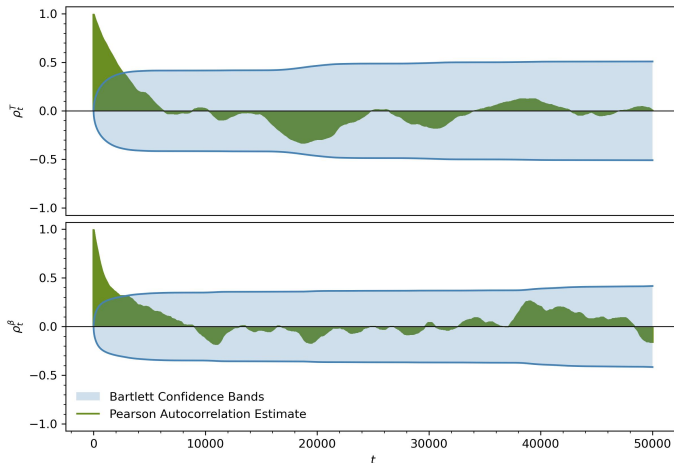


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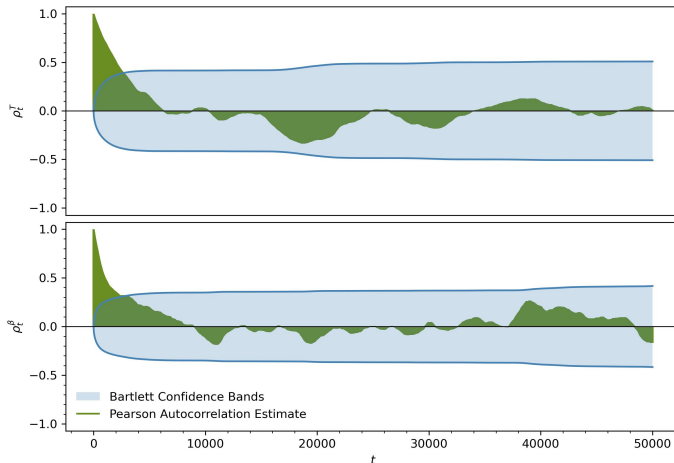
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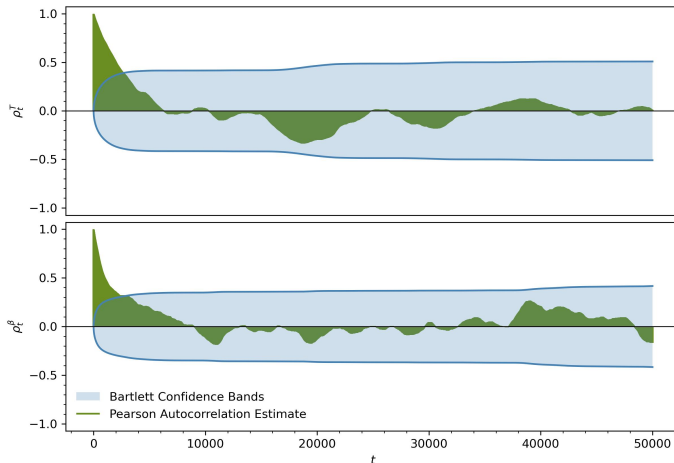
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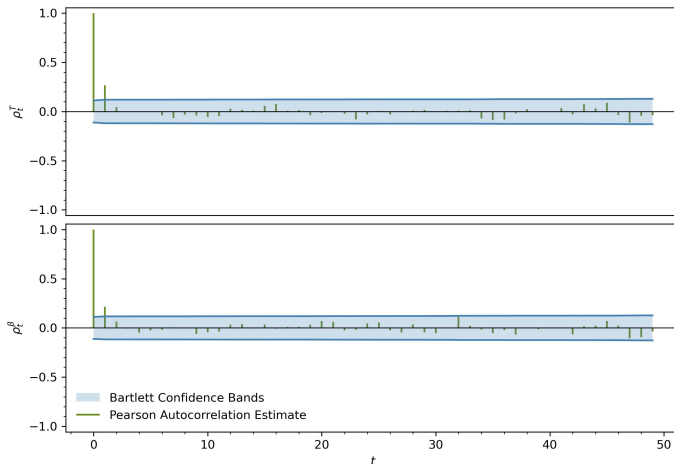
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- $h = 10^{-4}$ & acc = 97.72%
- acceptance too high
- step size too short
- $l \approx 3000$



Application

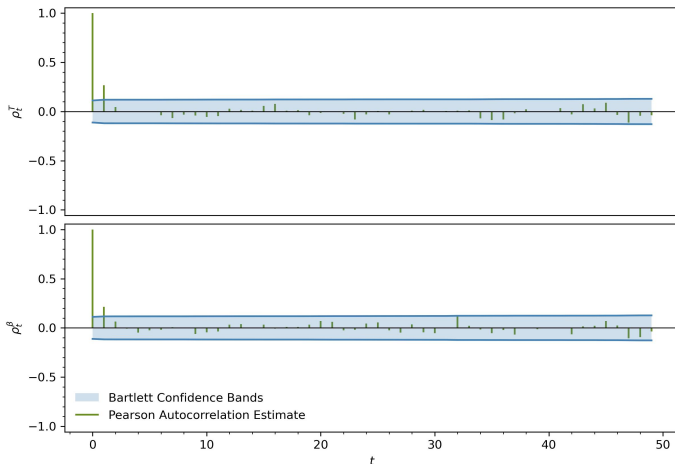
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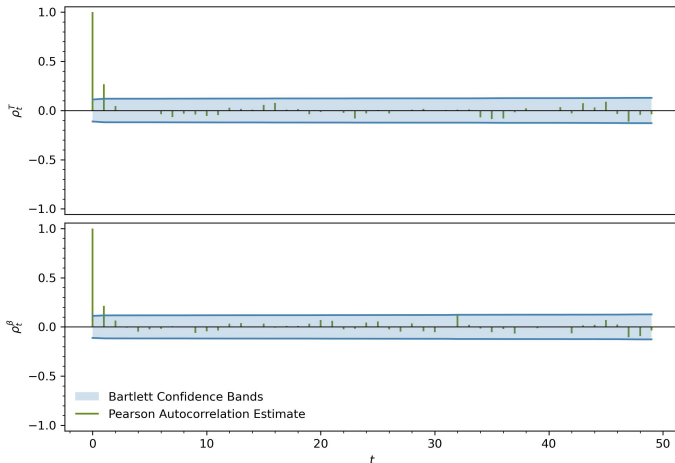


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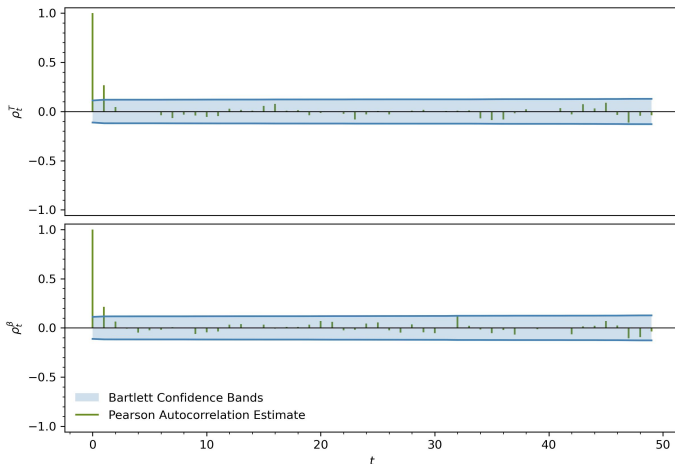
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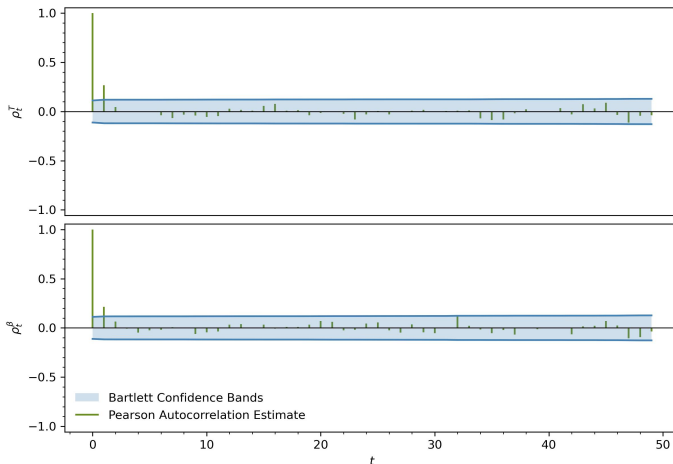
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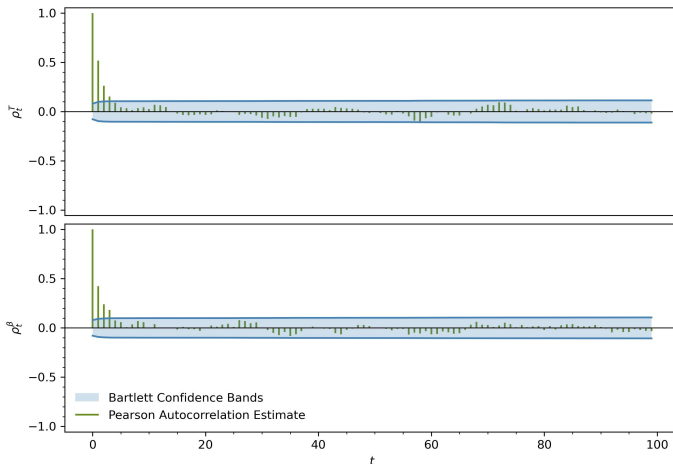
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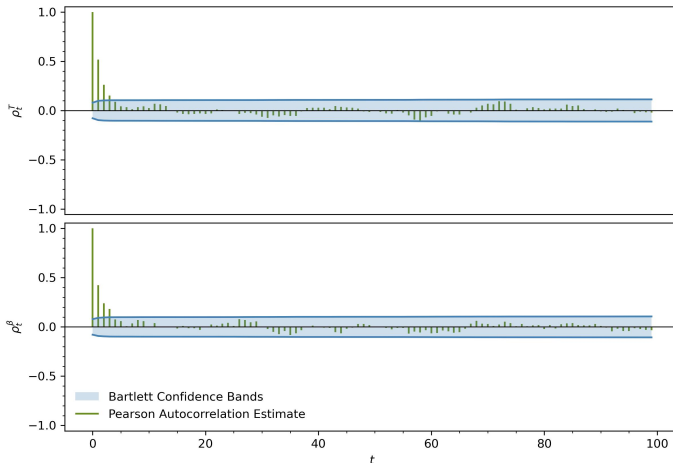
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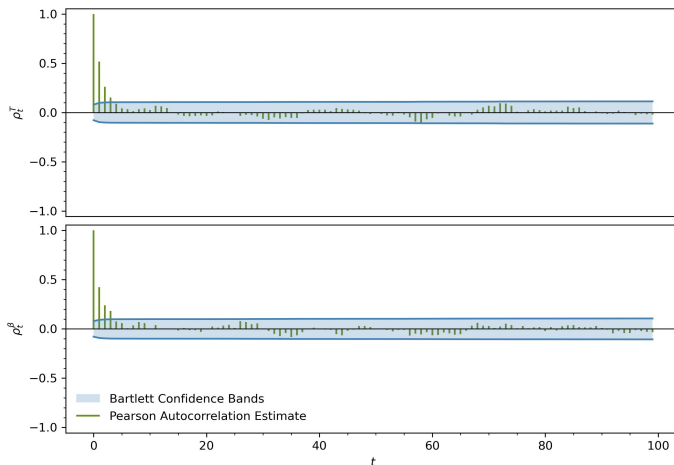


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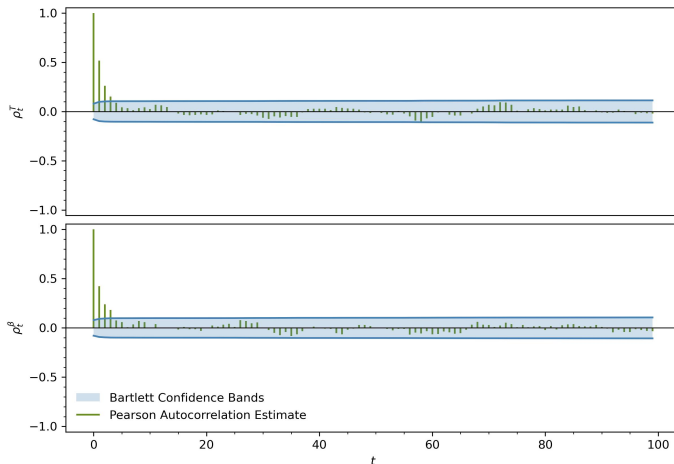
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 - *Importance Sampling*: weight samples from simple density to match target

Thank You!

- A. Casey, *Data Analysis and Machine Learning*³ 2022
- M. Linhoff & W. Rhode, *Statistical Methods of Data Analysis* 2023
- *emcee documentation*⁴ 2025

Resources on GitHub: [here](#)